Albert-Ludwigs-Universität Freiburg

Bachelor Project

Indoor Localisation via Direct and Reflected Signals in 3D with Two Walls and One Sender

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Abstract

This Project studies the three-dimensional ILDARS (Indoor Localization via Directed and Reflected Sound signals) problem for one signal source and two reflecting walls. The objective is to reconstruct a geometrically consistent configuration with the source position and the geometry of the reflecting walls for the actual received signals using only the incoming direction vectors and the time differences of arrival (TDOAs) of the received signals, without knowledge of the source's emission time or position. Using the geometric properties of direct and reflected signal paths, the problem is transformed into a system of constraints on the unknown source and wall parameters. The presented approach combines geometric preprocessing with numerical optimization to identify a configuration consistent with the observed measurements. Further it clears the questions about solvability of the three-dimensional setup and the ambiguities that may arise when reconstructing the geometry.

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1 Introduction

We look at the ILDARS(Indoor Localization using Directional and Reflected Sound signals) problem under the specification of one two three senders walls and dimensions.

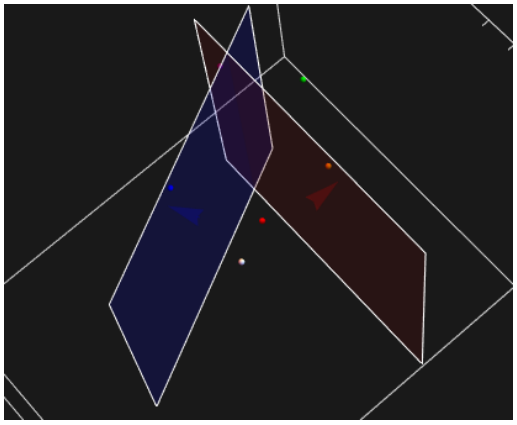


Figure 1: The Setup

The sender S and receiver R are randomly placed inside the same angle between the randomly placed walls, and then the sender S emits sound signals, which reach the receiver R either directly or through reflections off of the walls. This is also the only information the receiver R is getting, where it can label the signals with a timestamp, to further help in the localization later.

In Figure 1 we can see the setup of the localization problem. The receiver(white) is sat at at (0,0,0). Then there are the two walls w_1 and w_2 and the one sender(red)

randomly spread somewhere in the space, only with the condition that the sender S is placed in the same angular region as the receiver R between the walls.

2 Related Work

The relevant work for this bachelors project includes the early paper ”General ILDARS - What can a silent bat locate?”, 2023, by S.Mohanty and C. Schindelhauer[1] which the thesis ”Theoretical Analysis of the Two-Dimensional Indoor Localization Problem based on TDOA and Direction Vectors for one Source and two Reflecting Lines”, 2025, by Marcel Schuhmacher[2] buildes on, solving the exact scenario of one sender with two walls in 2D. The algorithm described in his thesis is part of the algorithm presented in this project for 3D.

Further, Luca Diringer also took on the specification of three walls with one sender in his thesis ”the Algebraic Reduction and Characterization of Equal Distance Point Pairs in Triangular Reflection Geometries”, 2026[3].

3 Background

The algorithm reconstructs a geometric configuration from time-difference-of-arrival (TDOA) measurements. Let p denote an unknown parameter, $tdoa_i$ the measured time differences, and $\mathbf{v}_i \in \mathbb{R}^3$ the given position vectors. The reconstructed points are defined by

$$\mathbf{s}_i = (p + tdoa_i)\mathbf{v}_i.$$

To determine p, planes are formed by using cross products such as $\mathbf{n}_1 = \mathbf{v} \times \mathbf{u}$ and $\mathbf{n}_2 = \mathbf{v} \times \mathbf{w}$ with for example

$$\mathbf{v} = (p + tdoa_0)\mathbf{v}_0 - (p + tdoa_1)\mathbf{v}_1.$$

The parameter p is determined by finding the configuration in which the corresponding planes satisfy the required geometric condition.

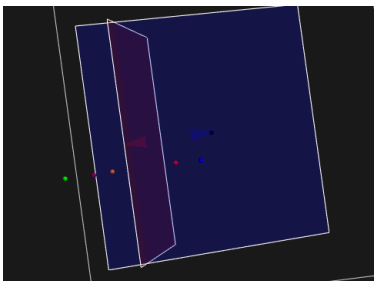


Figure 2: Sideview of the points plane

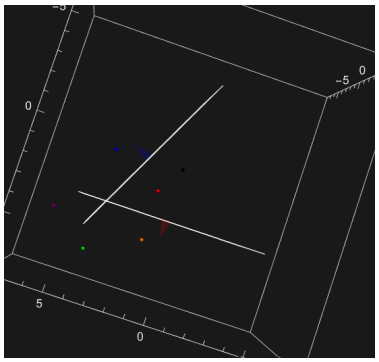


Figure 3: Topview of the points circle

The points $\mathbf{s}_i$ define a circle in a plane. Its center $\mathbf{c}$ is obtained from the perpendicular bisectors[4] of the point pairs, and its radius is

$$r = \|\mathbf{s}_0 - \mathbf{c}\|.$$

To represent the geometry in two dimensions, two orthogonal unit vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ are chosen in the circle's plane. Each point is projected onto this basis:

$$x_i = (\mathbf{s}_i - \mathbf{c}) \cdot \mathbf{e}_1, \qquad y_i = (\mathbf{s}_i - \mathbf{c}) \cdot \mathbf{e}_2.$$

The angular position of each point is then calculated using

$$\theta_i = \operatorname{atan2}(y_i, x_i),$$

with angles wrapped to the interval $[-\pi, \pi]$.

Finally, the parameters ϵ, ω, and ρ are adjusted through an optimization procedure to minimize the angular or geometric error E. The overall procedure combines vector geometry, planar projection, angular representation, and numerical optimization to obtain a consistent reconstruction.

This last part of the algorithm starting with representing the position of each point angularly, is the algorithm from M. Schuhmacher's thesis[2].

Note, that the specific setup this project addresses, includes one direct and three arbitrary signals.

4 Approach

The proposed approach reconstructs a geometric configuration from four signals with direction vectors and time-difference-of-arrival (TDOA) measurements. The central idea is to combine algebraic calculations with geometric constraints. First, possible values of an unknown parameter are calculated. These candidates are then evaluated using a circle-fitting criterion. Finally, the angular parameters are refined by minimizing the difference between measured and predicted point positions.

4.1 Problem Definition

TDOA-based reconstruction is useful in applications such as localization and signal processing, where measurements can be used to infer unknown geometric relationships. However, measurement uncertainty and geometric ambiguities can lead to multiple possible solutions. Therefore, the algorithm aims to identify a geometrically consistent configuration.

Let $\mathbf{v}_i \in \mathbb{R}^3$, for $i = 0, \ldots, 3$, denote four given direction or position vectors, and let $tdoa_i$ denote their corresponding TDOA measurements. The unknown scalar parameter p is used to construct four three-dimensional points:

$$\mathbf{s}_i = (p + tdoa_i)\mathbf{v}_i, \qquad i = 0, \ldots, 3.$$

The objective is to determine a suitable value of p and subsequently estimate the angular parameters describing the configuration.

The algorithm consists of two main stages. In the first stage, algebraic and geometric conditions are used to obtain and evaluate candidate values for p. In the second stage, the reconstructed points are represented in a two-dimensional coordinate system and an optimization procedure is applied to estimate the angular parameters.

4.2 Algebraic Computation of the Parameter p

The first part of the algorithm determines possible values of p by constructing two planes from the four input vectors. The planes are represented through their normal vectors. For the correct position of the signals S_i, the geometric condition is that the two plane normals are parallel, which can be expressed by the cross product being zero.

For a given parameter p, define the vectors

$$\begin{aligned}\mathbf{v}(p) &= (p + tdoa_0)\mathbf{v}_0 - (p + tdoa_1)\mathbf{v}_1,\\ \mathbf{u}(p) &= (p + tdoa_0)\mathbf{v}_0 - (p + tdoa_2)\mathbf{v}_2,\\ \mathbf{w}(p) &= (p + tdoa_0)\mathbf{v}_0 - (p + tdoa_3)\mathbf{v}_3.\end{aligned}$$

The normal vectors of the two planes are then calculated as

$$\mathbf{n}_1(p) = \mathbf{v}(p) \times \mathbf{u}(p), \qquad \mathbf{n}_2(p) = \mathbf{v}(p) \times \mathbf{w}(p).$$

Parallelism of the normals requires

$$\mathbf{n}_1(p) \times \mathbf{n}_2(p) = \mathbf{0}.$$

Since this is a vector equation, its three components are solved separately for p. The

resulting candidate solutions are restricted to positive values. Solutions obtained from the three components are then compared, allowing a small numerical tolerance, to identify consistent candidates.

If more than one candidate remains, the algorithm evaluates their circle charactristic to narrow the candidates down to only one.

4.3 Two-Dimensional Projection and Angular Optimization

After determining the one correct value for p, the reconstructed points are mapped into a two-dimensional coordinate system in the circle's plane. Neccessary for that are the normal vector $\mathbf{n}$ and the circle center M computed via the intersection of the perpendicular bisectors

$$b_1 = m_1 + ((S_1 - S_2) \times \mathbf{n})$$

and

$$b_2 = m_2 + ((S_1 - S_3) \times \mathbf{n})$$

solving

$$b_1 = b_2.$$

Denoted by m_1 and m_2 are the midpoints

$$m_1 = \left(\frac{(S_{1x} - S_{2x})}{2}, \frac{(S_{1y} - S_{2y})}{2}, \frac{(S_{1z} - S_{2z})}{2} \right)$$

and

$$m_2 = \left(\frac{(S_{1x} - S_{3x})}{2}, \frac{(S_{1y} - S_{3y})}{2}, \frac{(S_{1z} - S_{3z})}{2} \right)$$

used as support vectors for the bisectors.

The new origin for the two dimensional system is the point closest to $(0, 0, 0)$ in the plane, computed by

$$\mathbf{o} = (M.n) \cdot n.$$

Let $\mathbf{e}_1$ and $\mathbf{e}_2$ be orthogonal unit vectors spanning this plane. The projected coordinates of each point are

$$x_i = (\mathbf{s}_i - \mathbf{o}) \cdot \mathbf{e}_1,$$

$$y_i = (\mathbf{s}_i - \mathbf{o}) \cdot \mathbf{e}_2.$$

The corresponding angular position is then calculated using

$$\theta_i = \operatorname{atan2}(y_i, x_i).$$

Angles are wrapped to the interval $[-\pi, \pi]$ to handle the discontinuity at the angular boundary.

The final stage estimates the parameters ω, ϵ, and ρ. The Mathematica implementation models the angular position of point i as

$$\widehat{\theta}_i = \rho + 2\epsilon k_i - 2\omega b_i,$$

where b_i and k_i are the corresponding index or measurement values used by the algorithm. The predicted two-dimensional point is then

$$\widehat{\mathbf{q}}_i = \begin{pmatrix} M_x + r\cos(\widehat{\theta}_i) \\ M_y + r\sin(\widehat{\theta}_i) \end{pmatrix},$$

where $\mathbf{M} = (M_x, M_y)$ is the circle center in the projected coordinate system.

The parameters are estimated by minimizing the sum of squared coordinate differences between the predicted and reconstructed points:

$$E(\omega, \epsilon, \rho) = \sum_i \left\| \widehat{\mathbf{q}}_i - \mathbf{q}_i \right\|^2.$$

Mathematica's `FindMinimum` is used to perform this optimization from suitable initial values, subject to the parameter constraints implemented in the notebook. The resulting values provide the final angular description of the reconstructed configuration.

5 Experiments

This chapter evaluates the proposed reconstruction algorithm through a series of experiments. The experiments investigate the accuracy of the reconstructed geometry, the stability of the estimated parameters, and the influence of measurement errors. The evaluation is based on the geometric reconstruction and the optimization procedure described in Chapter 4.

5.1 Experiment 1: Reconstruction Accuracy

The first experiment evaluates whether the algorithm can recover a known geometric configuration. We use generated input data from a chosen configuration, allowing the reconstructed points and parameters to be compared with their reference values.

The reconstruction error is measured using the Euclidean distance between the estimated and reference points:

$$E_{\text{pos}} = \frac{1}{N} \sum_{i=0}^{N-1} \|\widehat{\mathbf{s}}_i - \mathbf{s}_i\|_2 \,.$$

The example configuration looks like this:

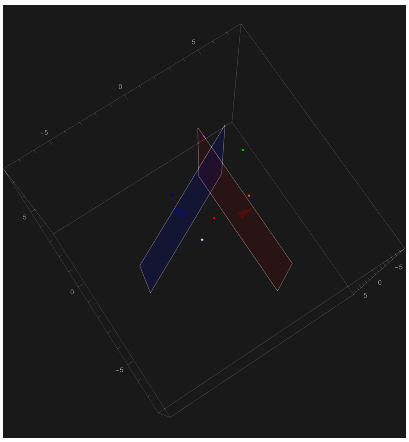


Figure 4: Example configuration 1

The corresponding input data for the first 4 needed signals the receiver acquires is the following:

Signal	**Direction (unit vector)**	**TDOA**
S_1	$\{0.733,\ 0.504,\ 0.458\}$	0
S_2	$\{0.0396,\ 0.273,\ 0.961\}$	1.852
S_3	$\{0.947,\ 0.237,\ 0.215\}$	2.460
S_4	$\{0.823,\ 0.155,\ 0.547\}$	4.913

Table 1: Signals from the sender reaching the receiver

To compare the results, the searched for value of p, circle center M and walls normal vectors n_{w1} and n_{w2} are the following:

$$p = 2.18403,$$

$$M = (3, 1.1, 3.5),$$

$$n_{w1} = (-0.4472, 0, 0.8944)^\top,$$

$$n_{w2} = (1, 0, 0)^\top.$$

Applying the algorithm returns a reconstructed geometry with the error

$$E = 6.52157 \cdot 10^{-18},$$

which suggest to be at least a correct geometry, if not the exact replication.
Looking at intermediate values, we can see that with

$$p = 2.18403,$$

and

$$M = (3, 1.1, 3.5),$$

the computation is exact, returning the correct true sender too. Taking a look at the walls, rebuilt through the angle optimization, we notice the computation leaves us the nomalized normal vectors

$$n_{w1} = (-0.4472, 0, 0.8944)^\top,$$

$$n_{w2} = (1, 0, 0)^\top,$$

proving itself as the initial conifguration.

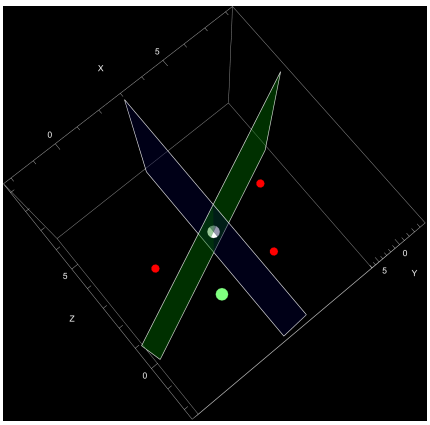


Figure 5: Reconstruction of example configuration 1

5.2 Experiment 2: Influence of Noise

This experiment investigates the sensitivity of the algorithm to perturbations in the input measurements.
As example configuration we choose the example configuration 1, as seen in Figure 4. Noise is added to the direction values values v_i according to

$$tdoa_i^{\text{noisy}} = tdoa_i + \eta_i,$$

where η_i represents a measurement perturbation. The algorithm is then executed using the perturbed measurements. The noise level chosen for this experiment is gaussian noise with a expected value of 0.02.

Using these turbulated direction data for the signals, the algorithm finds a configuration with an error of

$$E = 0.000378467.$$

This is an accepted geometry, since the error lies below the threshold of ε^{-3}, but in comparison to usual errors, i.e. the error in experiment 1, it is very high.
Again looking at intermediate values, the algorithm computes

$$p = 2.0959,$$

and

$$M = (2.98001, 1.06594, 3.45619).$$

These values show the impact of the perturbed data, hinting at reconstruction different to the initial configuration.
The returned walls normals are

$$n_{w1} = (0.62036, -0.01172, -0.78423)^\top,$$

and

$$n_{w2} = (0.97652, -0.00141, 0.21541)^\top.$$

All in all we have undeniable difference, but it is still acceptably low, looking at a plot of the reconstruction:

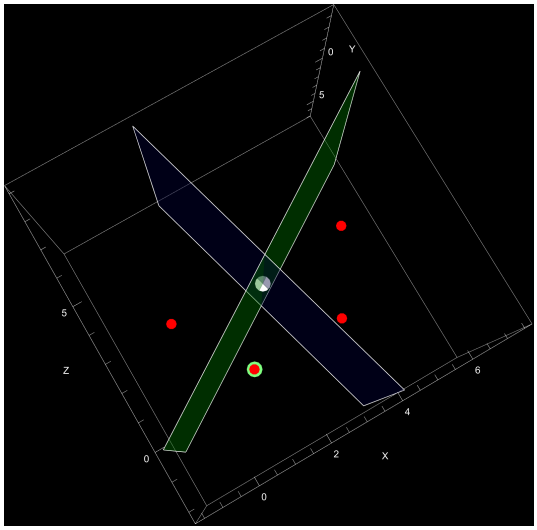


Figure 6: Reconstruction under noise

5.3 Experiment 3: Arbitrary configuration

The question under consideration in this experiment is about the ambiguity of the reconstruction.

We now randomly generate a setting through the inbuilt Wolfram Mathematica function `RandomReal[]`.

The generated points are:

$$S_1 = (0.824422, 0.111259, 0.446908), S_2 = (2.69353, 1.64867, 2.42514),$$

$$S_3 = (6.14423, 1.0125, 3.93894), S_4 = (3.31596, 1.75412, 2.83372).$$

The returned values for the walls normals are:

$$(0.597963, 0.491845, 0.632873)^\top,$$

and

$$(0.827723, 0.140225, 0.543334)^\top.$$

And the data reaching the receiver looks like this:

Signal	**Direction (unit vector)**	**TDOA**
S_1	$\{0.873014,\ 0.117817,\ 0.473249\}$	0
S_2	$\{0.676465,\ 0.414054,\ 0.60906\}$	3.03744
S_3	$\{0.833873,\ 0.137412,\ 0.534578\}$	6.42397
S_4	$\{0.705325,\ 0.373111,\ 0.602748\}$	3.75699

Table 2: Incoming data for the arbitrary setting

A Plot looks like this:

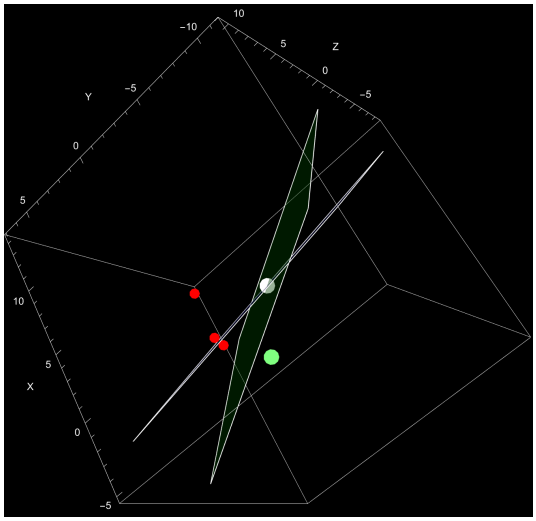


Figure 7: Arbitrary configuration

The algorithm returns a configuration with an error of

$$E = 3.98123 \cdot 10^{-16}.$$

The found configuration fits well. But lets take a look at all the computed values. The algorithm found the correct p and center:

$$p = 0.94434$$

$$M = (4.6653, -2.24422, 1.11801)$$

The main difference are the computed walls, therefore the found angles.

$$n_{w1} = (-2.72392, -2.24052, -2.88294)^\top,$$

and

$$n_{w2} = (-1.17488, -3.42901, -2.75908)^\top.$$

The reconstruction returns this configuration, classifiable as different by bare eye.

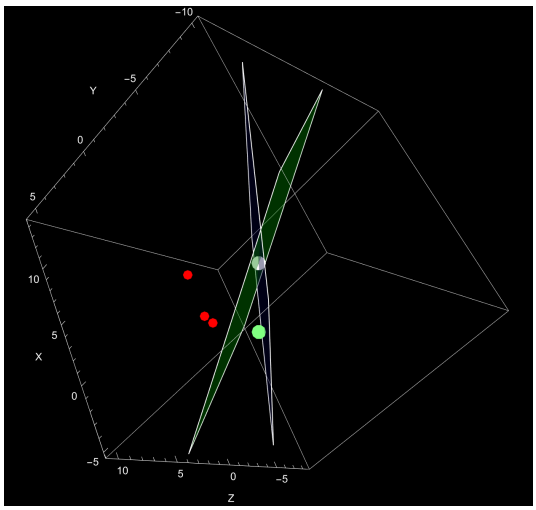


Figure 8: Reconstruction of arbitrary configuration

Yet this does not mean that the algorithm failed, it just simply found this different solution, which is a sound geometry, as better fitting.

5.4 Experiment 4: How many p are there?

The first stage may produce multiple candidate values for p. This experiment evaluates how frequently that happens, where the maximum in number of candidates lies and how differentiable the candidates are, in order to decide for the correct p.

For the differentiation between possible candidates, an error is computed, comparing the candidates by the circle characteristic in the following way:

- Find the circle center M for three of the four signals
- Measure the radius r
- Square the difference in radius for the fourth point

I have tested over a range of 1.000.000 generated configurations, resulting in a percentage of $\approx 6.56\%$ returning two candidates for p.

The limit of how many candidates for p exist lies practically speaking at two, but considering the 4 solutions for the quartic equation[5], it could be as high as four.
I have not found a case, where i could record more than two candidates for p, but neither this does, nor i can prove the non-existence.

Considering an example where the algorithm returns 2 candidates for p, we can look at the following input data:

Signal	**Direction (unit vector)**	**TDOA**
S_1	$\{0.3983, 0.725297, 0.561517\}$	0
S_2	$\{0.494859, 0.744995, 0.447323\}$	0.317516
S_3	$\{0.48384, 0.736329, 0.472989\}$	0.629306
S_4	$\{0.490456, 0.74512, 0.451938\}$	0.268951

Table 3: Input data returning 2 possible p

The computed candidates for p with error E are

$$p_1 = 1.00449, E = 1.11022 \cdot 10^{-15},$$

and

$$p_1 = 1.17574, E = 0.000142036.$$

The difference is common for a second value for p, making the decision process easy, although both yield a valid geometry. In no recorded case was the error similarly low.

6 Conclusion

This work investigated a geometric reconstruction algorithm based on time-difference-of-arrival (TDOA) measurements. The approach combines algebraic calculations, geometric constraints, and numerical optimization to estimate the unknown parameter p and reconstruct the corresponding configuration.

The experiments show that a minimum of four signals is still required for the reconstruction. Furthermore, the results indicate that extending the approach from two dimensions to three dimensions does not lead to a substantial difference in the overall behavior of the algorithm. This suggests that the main challenges observed in the reconstruction are not solely caused by the dimensionality of the problem.

A notable finding is that two possible values of p occur relatively frequently. The observed probability of this case is approximately 6.56%. In some of these situations, both candidate values produce reconstruction errors below the predefined threshold. Consequently, the attribute of the points plane alone is not always sufficient to distinguish between a solution.

In conclusion, the results demonstrate that the proposed approach is capable of performing the reconstruction, while also revealing limitations related to solution ambiguity. Future work should focus on evaluating whether additional geometric constraints or a more robust optimization strategy can improve the reliability of the algorithm, and maybe try finding a way without using the direct signal.

References (Literaturverzeichnis)

[1] S. Mohanty and C. Schindelhauer, "General ildars - what can a silent bat locate?," 2023.

[2] M. Schuhmacher, "Theoretical analysis of the two-dimensional indoor localization problem based on tdoa and direction vectors for one source and two reflecting lines," *Bachelors thesis*, 2025.

[3] L. Diringer, "Algebraic reduction and characterization of equal distance point pairs in triangular reflection geometries," *Bachelor thesis*, 2026.

[4] C. Maths, "How to find the center of a circle — definition, formula examples," *https://www.mathwords.com/c/construct$_c$irclecenter.htm*, 2026.

[5] "Quartic equation," *https://en.wikipedia.org/wiki/Quartic$_e$quation*, 2026.